Clinical Decision Support System for Disease Risk Assessment

Progress Report - 2

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Course: **CSIS 4495 – Applied Research Project**

Section: **03**

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1. Work Date / Hours Log

Date	Hours	Description of Work Done
Feb 25, 2026	3.0	Structured Kaggle diabetes dataset in folder. Implemented dataset cleaning logic (0 → NaN conversion, median imputation) and validated data consistency.
Feb 26, 2026	3.0	Conducted Exploratory Data Analysis including summary statistics, distribution plots, boxplots for outlier validation, and correlation matrix generation. Interpreted correlation values for feature relevance.
Mar 1, 2026	4	Implemented full 8-feature Logistic Regression and Random Forest baseline models. Evaluated accuracy, precision, recall, F1-score, ROC-AUC and documented performance comparison.
Mar 2, 2026	4	Implemented 5-fold Stratified cross-validation for reduced feature models. Performed threshold optimization analysis (0.50–0.30) and selected deployment threshold based on recall–precision tradeoff.

2. Summary Description of Work Done During This Reporting Period

In this reporting period, I worked on the enhancement of the data validation and baseline modeling elements of the disease risk assessment system. I prepared and cleaned the Kaggle diabetes data set by treating clinically invalid zero values and imputing the median.

EDA was performed to validate the feature distributions, check on the outliers and determine the correlations with the outcome variable. The decisions made were supported with correlation results (Glucose r=0.49, BMI r=0.31, Age r=0.24, BloodPressure r=0.16).

I used the entire 8-feature baseline models (Logistic Regression and Random Forest) and tested their predictive abilities. In order to increase methodological rigor, I included 5-fold Stratified cross-validation to determine model stability. Also, I performed threshold optimization to enhance screening-related recall and chose threshold 0.45 as a balance that seemed clinically appropriate.

The basic project proposal was not altered, but the evaluation procedure was enhanced following the feedback during the mid-term.

3. Repo Check-In of Implementation Completed

Since the last progress report, I have checked in multiple updates under the Implementation folder related to dataset validation and model evaluation. My commits include the cleaned dataset processing logic (common.py), exploratory data analysis script (eda_analysis.py), and the full 8-feature baseline modeling script. I also did the cross-validation implementation and threshold analysis code used to evaluate recall–precision trade-offs. The corresponding output artifacts (cross-validation summary, threshold analysis CSV, and evaluation metrics) were saved under the outputs/metrics directory.

The following files and artifacts were committed under my contributions:

- modeling/common.py (data cleaning and evaluation utilities)
- modeling/eda_analysis.py
- modeling/train_full_8_features_uthara.py
- Cross-validation outputs under outputs/metrics
- Threshold analysis CSV and evaluation artifacts

4. AI Use Section

AI Tool	Version	Purpose of Use	How It Added Value
ChatGPT	GPT-5 (Premium)	Help with implementing 5-fold cross-validation in sklearn	Helped me understand how to structure cross-validation properly. I wrote and tested the code myself and verified the ROC-AUC results.
ChatGPT	GPT-5 (Premium)	Help with testing different classification thresholds	Helped me understand how to check precision and recall at different probability levels. I implemented the threshold loop and selected 0.45 based on the results.

5. APPENDIX

5.1 Appendix A – AI Prompts

1. How do I implement 5-fold Stratified cross-validation in sklearn for a binary classification model?
2. How can I test model performance at different probability thresholds to improve recall

5.2 Appendix B – Cross-Validation Stability

5-Fold Stratified Cross-Validation (Reduced 4 Features)

Logistic Regression:

- Mean ROC-AUC: 0.825
- Std Dev: 0.030

Random Forest (Baseline):

- Mean ROC-AUC: 0.812
- Std Dev: 0.041

This confirmed model stability beyond single-split evaluation.

5.3 Appendix C – Threshold Optimization Study

Tuned Random Forest evaluated at multiple thresholds:

Threshold	Precision	Recall	F1
0.50	0.597	0.796	0.683
0.45	0.595	0.815	0.688

Threshold 0.45 selected to increase sensitivity for screening while maintaining balanced predictive performance.